KAREL KUTCHAN

Senior Software Engineer

Prague, Czechia | +420 605 570 494 | karel@codeguy.cz | LinkedIn | GitHub | codeguy.cz

PROFESSIONAL SUMMARY

Senior Software Engineer with more than ten years of commercial experience. Until April 2026 I was Lead Frontend Engineer at BlueGhost, owning frontend architecture, delivery standards and engineering quality across product teams; since February 2026 I design and build Seeder, a production Django and PostgreSQL system for the National Library of the Czech Republic.

My work centres on React and TypeScript applications that have to last: component foundations, accessibility, testing strategy and code review, with hands-on CMS and platform background across Kentico, Kontent.ai, Payload CMS and PHP/Twig. I keep learning deliberately as a core student at 42 Prague, and use AI-assisted workflows while keeping architecture and maintainability under engineering control.

CORE SKILLS

Frontend engineering	React 18/19, TypeScript, Next.js, JavaScript, HTML5, CSS3, functional React patterns
State and data	TanStack React Query, Zustand, Redux, MobX, URL-synced UI patterns, typed API integration
Validation and forms	Configuration-driven UI, reusable form fields, schema-aware validation, sync and async validation patterns
CSS and UI systems	SCSS/Sass, CSS Modules, styled-components, LESS, specificity and cascade, responsive layouts, design systems, Storybook
CMS and platform integration	WordPress, Kentico CMS, Kontent.ai, Payload CMS, PHP/Twig, monorepo-ready workflow
Quality and tooling	Vitest, Playwright, React Testing Library, Git, PNPM/NPM/Yarn, code reviews, CI/CD practices, AI-assisted development workflows
Accessibility and localization	React Aria hooks, keyboard and screen-reader usability, WCAG-oriented implementation, internationalisation-aware UI, RTL-aware CSS

EXPERIENCE

Software Engineer | Národní knihovna ČR | 02/2026 – Present

Stack: Python, Django, PostgreSQL, Docker, Playwright, Git

- Seeder is a production Django application backed by PostgreSQL. Curators use it to register web sources and publishers, track licensing agreements, schedule harvests, and assemble thematic collections for webarchiv.cz.
- I design and build across the whole application: the PostgreSQL data model, Django logic and integrations, the admin interface and the workflows curators use every day. I take changes from the database all the way to the screen.
- Django and PostgreSQL are the core. To deliver changes safely I use Playwright tests, pull requests and Docker, then deploy them through Jenkins to the test and production environments.

Lead Frontend Engineer | BlueGhost | 03/2025 – 04/2026

Stack: Next.js, React, TypeScript, PNPM, TanStack React Query, Zustand, SCSS Modules, Storybook, React Aria, Vitest, Playwright, Git

- Architected a reusable React/Next.js frontend foundation for web platforms by defining application structure, component boundaries and TypeScript conventions, improving long-term maintainability across current and future projects.
- Improved release confidence by combining Storybook-driven component workflows with Vitest/Playwright testing practices and stricter review standards instead of relying only on manual QA.
- Standardized server-state and client-state patterns with TanStack React Query and Zustand, making data loading, cache invalidation and UI state easier to reason about in feature development.
- Raised frontend quality by leading code reviews, mentoring engineers and enforcing conventions around strict TypeScript, SCSS Modules, reusable UI patterns, accessibility and performance.
- Introduced structured AI-assisted development workflows — Copilot, Gemini/Codex-style CLI workflows, system prompts, Figma-to-token/code assistance — while keeping architecture, code review and maintainability under engineering control.

Frontend Engineer | Kontent.ai | 06/2024 – 02/2025

Stack: React, TypeScript, Next.js, styled-components, React Aria, Storybook, Playwright, Git

- Improved accessibility of a complex React application by refactoring UI behavior with React Aria hooks and custom accessible interaction patterns for keyboard and screen-reader users.
- Reduced accessibility risk by reviewing landmarks, labels, focus behavior and component semantics against strict usability requirements instead of treating a11y as a final QA step.
- Kept refactoring safe by collaborating with the team on maintainable TypeScript components, Storybook workflows and Playwright-based checks in a contract environment with defined delivery scope.

Lead Frontend Engineer | TLDR;IT s.r.o. | 04/2024 – 09/2024

Stack: Next.js, React, TypeScript, Sass, CSS Modules, Payload CMS

- Delivered frontend architecture for time-boxed web and mobile-oriented projects by selecting scalable React/Next.js patterns, reusable component structures and CSS Modules/Sass styling strategy.
- Connected modern React interfaces to CMS-driven content using Payload CMS and TypeScript models, strengthening my ability to work with editorial/admin data structures and frontend contracts.

Frontend Developer | eMan a.s. | 09/2021 – 04/2024

Stack: React, Next.js, TypeScript, MobX, Redux, Sass, CSS Modules, Storybook, Git

- Built and evolved React/TypeScript interfaces for enterprise clients including E.ON, MND, ČEZ and Bohemia Energy, increasing reuse and consistency through design systems and component libraries.
- Built Moje MND, a customer self-service portal, from scratch in a two-person frontend team — forms, validation, tabular overviews, invoices, contracts and document handling.
- Delivered two internal E.ON applications: PUDplus for work orders, findings and preventive maintenance, and MKME web for work orders and metering.
- Improved maintainability of client applications by refactoring legacy frontend code into modern React/TypeScript patterns, reusable components and clearer styling architecture.
- Supported performance, accessibility and responsive UX improvements by working closely with UX and backend teams on production web applications used by large customer bases.

Frontend Developer | LMC | 09/2020 – 09/2021

Stack: React, JavaScript, PHP/Twig, Less, Sass

- Delivered custom web solutions for corporate clients on the jobs.cz ecosystem by translating client and design requirements into responsive frontend implementations.
- Worked comfortably in a PHP/Twig and frontend hybrid environment, strengthening the same PHP-to-frontend collaboration mindset needed for React features embedded in CMS platforms.

Frontend Developer | Amp X | 09/2019 – 09/2020

Stack: React, TypeScript, Redux, JSS, Storybook, Git

- Built React/TypeScript UI for a distributed-energy platform where data clarity and reliable interaction patterns were critical for market-facing workflows.
- Maintained readable and sustainable frontend code with Redux, Storybook and component-based UI practices in a product development team.

Frontend Developer | **Skype.com** | 04/2019 – 09/2019

Stack: JavaScript, jQuery, HTML5, CSS3, Sass, Kentico CMS, Git

- Contributed to a high-traffic public website serving more than 400,000 unique users daily across 30+ countries by improving responsive components, accessibility and performance.
- Worked inside a CMS-backed publishing environment with Kentico, aligning frontend components with editorial/content-management workflows on a global public site.

Frontend Coder | **Foxconn DRC s.r.o.** | 06/2018 – 03/2019

Stack: HTML5, CSS3, Sass, React, BEM, Git

- Developed responsive interfaces and a BEM-based component foundation.

Frontend Coder | **Mountfield, a.s.** | 05/2017 – 05/2018

Stack: HTML5, CSS3, JavaScript, jQuery, Kentico CMS

- Maintained the main website and delivered supporting marketing and internal interfaces.

Web Administrator, Graphic Designer, and Junior Frontend Coder | **BitWare CZ s.r.o.** | 08/2014 – 05/2017

Stack: HTML5, CSS3, JavaScript, jQuery, PHP, WordPress

- Worked across website administration, visual design and early frontend delivery.

Earlier experience: Sales Manager · Austro-Bohemia (2011–2014) — CRM/ERP and ICT services.

EDUCATION & LANGUAGES

42 Prague (2026–Present)
Core student — software engineering fundamentals in C and Python

COPTH (2002–2006)
Electrical engineering, telecommunications and computer technologies

Czech — Native

English — Working professional

Swedish — Beginner